

A computer-assisted all-subcritical Birkhoff theorem at a fixed mass ratio for the planar circular restricted three-body problem

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Abstract

We give a computer-assisted proof of a Birkhoff global-section conclusion for the planar circular restricted three-body problem at one exact mass parameter and for every subcritical energy. The fixed auxiliary parameter is $\gamma = 459/1000$. We use the Jacobi convention $\Sigma_c = H^{-1}(-c)$ and write $\Delta = c - c_{L_1}(\gamma) > 0$ for the subcritical energy distance from the first critical value. The proof has two complementary regimes. For $\Delta \geq 3/50000$, an explicit exact global momentum shear generated by $G(q) = -(53/250)q_1q_2$ converts the regularized double covers into strictly convex hypersurfaces; exact symbolic reduction, interval/Bernstein certification, and an independent MPFR-192 replay establish the required positivity. For $0 < \Delta < 3/50000$, we use a corrected two-component Levi-Civita description, a segmentwise rotation/Conley–Zehnder estimate, finite validated return geometry, and a first-hit decomposition of the only complementary β -bad block. Two non-pointwise seed cells are discharged by positive integrated entry-plus-residence margins rather than by a false uniform β bound. The complete theorem-critical finite certificate families have independent R16 replay/checker support at the project’s I2 level or stronger where stated. The resulting regularized Reeb flows are dynamically convex. In each bounded component, a standard Birkhoff retrograde periodic orbit then binds a rational open book whose disk-like pages are global surfaces of section.

Keywords: restricted three-body problem; global surface of section; dynamical convexity; Conley–Zehnder index; computer-assisted proof; interval arithmetic.

1 Introduction

The planar circular restricted three-body problem (PCR3BP) is a fundamental two-degree-of-freedom Hamiltonian model in celestial mechanics. Below the first critical Hamiltonian energy, the Hill region has two bounded components around the primaries. After collision regularization, each physical component is naturally described as an $\mathbb{R}P^3$ -type energy hypersurface whose double cover is diffeomorphic to S^3 .

Birkhoff’s global-section problem asks for a disk-like global surface of section bounded by a distinguished retrograde periodic orbit; see (Birkhoff, 1915). Modern contact and symplectic methods reduce this question to dynamical convexity. In the convention used here, a regularized contact form is dynamically convex when every contractible periodic Reeb orbit has lower-semicontinuous disk-trivialized Conley–Zehnder index at least three. Strict convexity of an S^3 -type energy hypersurface in $\mathbb{R}^4$ is a classical route to this property through the work of Hofer, Wysocki and Zehnder (Hofer et al., 1998).

The central issue in the present work is that one geometric mechanism does not remain uniformly effective all the way to the first critical level. We therefore combine two rigorous proof architectures. Far enough below the first critical Hamiltonian energy, the same explicit exact symplectic shear used in the fixed-energy G459 proof yields strict convexity and propagates

monotonically along the energy ray. In the thin near-critical collar, we replace global Euclidean convexity by an orbitwise but finite rotation-index argument. The near-critical proof is not an exhaustive enumeration of periodic orbits. It uses a finite cover, validated inequalities, a finite return graph, and an integrated first-hit compensation argument that controls the only two cells where a uniform pointwise $\beta \geq 1/100$ estimate is unavailable.

The result is a fixed-mass, all-subcritical theorem. It does not assert a mass interval and does not close the separate all-mass continuation program.

1.1 Exact parameter and main theorem

We fix

$$\gamma = \frac{459}{1000}.$$

The exact mass labeling used by the regularized Hamiltonian is

$$\mu_* = \frac{264377992475701}{441699674239000}, \quad 1 - \mu_* = \frac{177321681763299}{441699674239000}.$$

We use the Jacobi convention $\Sigma_c = H^{-1}(-c)$ and define

$$\Delta = c - c_{L_1}(\gamma).$$

Thus $\Delta > 0$ is the subcritical side.

Table 1: Exact parameter and convention data used throughout the proof.

Item	Exact convention/value	Publication role
Auxiliary mass parameter	$\gamma = 459/1000$	Fixed throughout; no mass interval is claimed.
Regularized mass labels	$\mu_* = 264377992475701/441699674239000$ and $1 - \mu_* = 177321681763299/441699674239000$	Exact rational reconstruction used by the parameter audit.
Energy convention	$\Sigma_c = H^{-1}(-c)$, $\Delta = c - c_{L_1}(\gamma)$	$\Delta > 0$ is the subcritical side.
Deep/near split	$\Delta_0 = 3/50000$	Strict-convexity propagation for $\Delta \geq \Delta_0$; rotation/return proof for $0 < \Delta < \Delta_0$.
Index convention	Lower-semicontinuous, disk-trivialized μ_{CZ} for contractible orbits	Dynamical convexity means $\mu_{CZ}^{\text{disk}} \geq 3$.
LC/time semantics	Two focus-specific Levi-Civita charts with positive time factors; reduced frame/trivialization transported with the cocycle	No global equal-clock identification is used.

Theorem 1.1 (Fixed-G459 all-subcritical dynamical convexity and global sections). *Fix $\gamma = 459/1000$. For every subcritical level $\Delta > 0$, each bounded regularized physical component has dynamically convex Reeb flow: every contractible periodic orbit has disk-trivialized lower-semicontinuous Conley–Zehnder index at least three. Consequently, in each bounded component, a Birkhoff retrograde periodic orbit in the standard topological/shooting class binds a rational open book whose pages are disk-like global surfaces of section.*

The topological quantifier in the second sentence is deliberate. We use a standard Birkhoff retrograde orbit with the 2-unknot and rational self-linking $-1/2$ properties appearing in the PCR3BP literature. We do not claim that every orbit informally described as retrograde automatically satisfies these knot-theoretic hypotheses.

1.2 Proof architecture

The proof splits the subcritical ray at the exact rational threshold

$$\Delta_0 = \frac{3}{50000}.$$

For $\Delta \geq \Delta_0$, we prove strict convexity of a sheared regularized S^3 lift. For $0 < \Delta < \Delta_0$, we prove the Conley–Zehnder lower bound directly from validated rotation and return geometry. The near-critical certificates may use the critical face $\Delta = 0$ as a boundary of an interval enclosure; the theorem itself concerns $\Delta > 0$.

2 Hamiltonian, regularization, and conventions

The sign convention $\Sigma_c = H^{-1}(-c)$ is fixed throughout. At the first critical point L_1 , $H(L_1) = -c_{L_1}$, so $\Delta > 0$ implies $-c < -c_{L_1}$.

The two bounded components require focus-specific Levi–Civita charts in the near-critical argument. The final proof uses the corrected two-component semantics: the two charts have positive Hamiltonian time changes, and reduced linearized cocycles are compared after transporting the reduced frame/trivialization. Equality of Conley–Zehnder and Robbin–Salamon data follows from the transported reduced-cocycle conjugacy under positive time reparametrization, not from equality of the time parameters.

3 Fixed-energy convexity anchor and deep-energy propagation

3.1 Exact global symplectic shear

At the reference G459 level, the regularized Hamiltonian admits the exact global momentum shear generated in physical coordinates by

$$G(q) = -\frac{53}{250}q_1q_2.$$

In elliptic-hyperbolic regularizing coordinates this is equivalent to

$$g(x) = \sinh(2x_1) \sin(2x_2) = 16q_1q_2, \quad \lambda = -\frac{53}{4000}.$$

Because the momentum translation is by an exact one-form, it is an exact symplectomorphism, and the generator is compatible with the deck involution.

3.2 Exact tangent-Hessian reduction

After the shear the Hamiltonian remains in the magnetic-mechanical class

$$H(y, x) = \frac{1}{2}|y + F(x)|^2 + V(x).$$

We use the generalized tangent-Hessian criterion of Liu and Salomão (Theorem 9.1 in the current arXiv version) (Liu and Salomão, 2025). Exact symbolic elimination of the fiber angle reduces the positivity problem to

$$L(s, t) = A_0(s, t) - \sqrt{R_1(s, t)} - \sqrt{R_2(s, t)} > 0,$$

where $s = \cosh x_1$, $t = \cos x_2$, and A_0, R_1, R_2 are exact rational bivariate polynomials.

The frozen fixed-energy certificate has 13,655 final leaves: 11,902 PASS, 1,753 OUTSIDE, and no PENDING leaf. The smallest certified lower bound is

$$8.331573711073198 \times 10^{-9} > 0.$$

A clean R14 replay regenerated the exact polynomial data, rebuilt GCC and Clang computations, reconstructed the final ledger byte-for-byte, independently audited the tiling, and replayed all 13,655 leaves with MPFR-192 directed rounding.

3.3 Rebased energy-ray propagation

The publication-critical base threshold is $\Delta_0 = 3/50000$. At this level the finite strict-convexity certificate closes, and exact energy-ray monotonicity propagates positivity for every $\Delta \geq \Delta_0$. The R16 independent replay checks all 15,364 global terminal leaves with MPFR-192 and zero failures, including the historical 4,207 expected-PASS residual leaves. Therefore all levels $\Delta \geq 3/50000$ are dynamically convex through strict convexity.

4 Near-critical rotation and finite return geometry

We now restrict to $0 < \Delta < 3/50000$.

4.1 Segmentwise rotation estimate

In the final transported reduced contact frame, the project constructs a pointwise quantity $\beta(w)$ satisfying

$$\Theta(b) - \Theta(a) \geq \int_a^b \beta(w(t)) dt.$$

The frozen rotation-index convention gives

$$\Theta(T) > 2\pi \implies \mu_{CZ}^{\text{disk}} \geq 3.$$

On the canonical near-critical beta-chart partition, the finite certificate closes all 32,661 parent obligations for the target $\beta \geq 1/100$ on its intended domain.

The same framework uses stronger regions where $\beta \geq 2/5$ and $\beta \geq 7/200$. Let t_{40} and t_{35} denote the certified occupation times in those two stronger regions, and let t_R be the remaining time to which the baseline bound $\beta \geq 1/100$ applies. Thus

$$T = t_{40} + t_{35} + t_R.$$

The exact occupation identity used by the frozen N2 checker is

$$\frac{2}{5}t_{40} + \frac{7}{200}t_{35} + \frac{1}{100}t_R = \frac{1}{100}T + \frac{39}{100}t_{40} + \frac{1}{40}t_{35}.$$

Consequently the segmentwise estimate gives

$$\Theta(T) \geq \frac{1}{100}T + \frac{39}{100}t_{40} + \frac{1}{40}t_{35}.$$

If a contractible periodic orbit had $\mu_{CZ}^{\text{disk}} \leq 2$, the frozen rotation-index implication forces $\Theta(T) \leq 2\pi$. Multiplication by 200 therefore gives the exact low-index occupation budget

$$2T + 78t_{40} + 5t_{35} \leq 400\pi.$$

The convention audit also checks the Robbin–Salamon/Conley–Zehnder comparison and the complementary contribution used in the underlying N2 bridge; no untransported LC frame or equal-clock shortcut enters this inequality.

4.2 Current-neck short-period exclusion

The current-neck residual is controlled by validated Yorke/Wirtinger-type period estimates, exact energy support, and displacement inequalities. The final theorem is

$$T_{\hat{H}} > 3$$

for every nonconstant periodic orbit represented by the canonical current-neck residual. The independent R16 replay reconstructs exactly the canonical 1,222 boxes: 1,212 close by the unweighted Yorke test, nine of the ten residual boxes close by the fixed-diagonal weighted Yorke test, and the single remaining index closes by the independent scalar-displacement certificate. Thus the accounting is 1,222/1,222, with no missing or extra canonical key.

4.3 Finite return graph and long regime

The residual return geometry is encoded by a finite θ -ordered graph. The independent N4 backend validates all 30/30 theorem-critical half-phase transitions. Its conservative collapsed hard graph has 34 hard edges, no self edge, no θ -order violation, is acyclic, and has maximum hard-edge path length 5. This is the property used downstream: a completed residual cluster contains at most seven residual visits.

The N5 replay then checks the validated propagation obligations that make the counting argument quantitative: 2,392/2,392 strong-window obligations, 2,362/2,362 outer safe half-obligations, 10/10 hard-sink half-obligations, and 2,372/2,372 detachment half-obligations pass. The exact residence estimate is

$$T_{\text{res}} < \frac{145\pi}{106496} < \frac{1}{230},$$

while the occupation argument gives $t_R \geq 219/100$. Hence a hypothetical long low-index orbit must make at least 504 residual visits. Since N4 allows at most seven visits per completed cluster, at least 72 clusters are required. The frozen exact cluster arithmetic contradicts the available long-regime occupation budget. This closes the hypothetical low-index regime $T > 400\pi/7$ without enumerating periodic orbits.

5 The complementary seed-7 block

The last local obstruction arises from an exact 16-cell seed cover. Fourteen cells satisfy the pointwise theorem $\beta \geq 1/100$. Two cells, denoted $s04_d0$ and $s06_d0$, do not. They are never relabeled as pointwise $\beta \geq 1/100$ cells; instead, their union is controlled by an integrated first-hit argument.

5.1 One-way bad block and corridor

Let $B = s04_d0 \cup s06_d0$. On B , and on a larger surrounding corridor, validated inequalities give

$$\dot{\theta} < 0, \quad \dot{A} < 0,$$

and the two ϕ faces point strictly outward. Hence local recirculation is impossible.

5.2 Theta-entry geometry

On the complete theta-entry shell, $\dot{A} < 0$. All possible negative β is confined to a narrow exact sublayer N , where

$$\beta \geq -\frac{3}{200}$$

and $\dot{A}$ is sufficiently negative to give

$$\beta - \frac{3}{20}\dot{A} > 0.$$

Inside B itself, $\beta \geq 0$ on both exceptional cells.

5.3 Upper-theta entry family

The complete upper-theta entry face has the exact natural partition

$$32 \times 32 \times 3 = 3072$$

roots. The finite genealogy closes all roots and yields

$$J_{001} > 0.005708670873993645.$$

The generic following- B deficit is smaller than 0.0049211744370937422, hence the paired upper-theta margin is

$$> 0.0007874964368999028 > 0.$$

5.4 Upper-A entry family

On the high- A corridor, $\beta \geq 1/20$, and every upper- A -reaching first-hit block spends more than $1/40$ of the relevant time there. Relative to the $\beta = 1/100$ target this yields

$$J_{001}(\text{entry}) > \left(\frac{1}{25}\right) \left(\frac{1}{40}\right) = \frac{1}{1000}.$$

The following B residence satisfies $T_{B,\text{upperA}} < 17/200$, so the deficit relative to $1/100$ is $< 17/20000$. Therefore

$$\frac{1}{1000} - \frac{17}{20000} = \frac{3}{20000} > 0.$$

Lemma 5.1 (First-hit decomposition of B). *Decompose an orbit into maximal connected residence intervals in B . Every forward entry into B occurs through exactly one of two families: the upper-theta face or the upper- A face. Pair each maximal residence with its immediately preceding first-hit entry block. These pairs are disjoint in time and no entry block is reused.*

Proof. The outward ϕ -face signs exclude forward ϕ entry. The strict θ and A monotonicities leave only the upper-theta and upper- A faces as forward entry faces. Maximal connected residences are pairwise disjoint by definition. The immediately preceding first-hit block terminates at the corresponding entry event and cannot simultaneously precede a distinct maximal residence. $\square$

Corollary 5.2. *Each entry-plus- B pair satisfies the effective $\beta = 1/100$ rotation budget. Hence the two exceptional seed cells are discharged by integrated rotation, and the exact seed-7 cover is closed $16/16$.*

6 Dynamical convexity and global surfaces of section

In the deep branch, strict convexity gives dynamical convexity by (Hofer et al., 1998). In the near-critical branch, assume a contractible periodic orbit had disk-trivialized $\mu_{\text{CZ}} \leq 2$. The corrected two-component semantics, segmentwise rotation inequality, current-neck exclusion, finite return graph, long-regime contradiction and complete seed-7 discharge reduce every

possible itinerary to one of the validated closed cases, contradicting the low-index assumption. Thus every contractible orbit has $\mu_{\text{CZ}}^{\text{disk}} \geq 3$.

The exact shear preserves disk-trivialized CZ under transported capping-disk trivializations. For the regularization double cover $S^3 \rightarrow \mathbb{R}P^3$, a contractible capping disk lifts, and the pulled-back contact planes and linearized flows give the same disk-trivialized index upstairs and downstairs.

The standard Birkhoff retrograde orbit on a subcritical regularized component has the 2-unknot and rational self-linking $-1/2$ topology used in the literature; the frozen source audit uses the discussion at Theorem 1.5 of Liu and Salomão (2025). Hryniewicz and Salomão prove the rational-open-book/global-section implication for a p -unknotted orbit with self-linking $-1/p$ in a dynamically convex Reeb flow on a lens space of order p ; in the frozen cited version this is Corollary 1.8 (Hryniewicz and Salomão, 2016). Taking $p = 2$ gives the final statement of Theorem 1.1.

7 Computer-assisted proof architecture and trust base

The proof distinguishes search/generation from theorem acceptance. Adaptive generators, numerical scouts, plots, ordinary floating integrations and LLM-produced text or code are not theorem authorities. Theorem-critical computational claims are accepted only through exact arithmetic, validated interval arithmetic, finite proof objects and fail-closed checkers.

The project independence scale is: I0, same-program rerun; I1, clean-environment or cross-compiler replay; I2, independent replay/checker of the same finite certificate; I3, independently derived checker or alternate arithmetic implementation; I4, independent researcher/team reproduction; I5, independent reconstruction from the paper specification. Every theorem-critical finite certificate family in the final G459 chain meets at least I2. I3 is claimed only where the implementation independence actually justifies it.

Table 2 gives the publication-facing traceability map. The release manifest records the full SHA-256 values for every frozen package, while the internal Claim–Evidence Matrix retains the finer claim-by-claim genealogy.

Table 2: Compact proof-to-evidence map for theorem-critical nodes.

Manuscript role	Nodes	Frozen R16 evidence	Acceptance
Parameter/fixed anchor	F0–F2	<code>R16_FIXED_TRUST_BASE_I2_EVIDENCE_20260926.zip</code> and external-theorem/convention audit	exact/I2 + theory audit
Deep-energy propagation	D1	<code>R16_D1_I2_GLOBAL_MPFR_REPLAY_20260925.zip</code>	15,364/15,364
LC semantics and rotation budget	N0–N2	N0 semantic audit; N1 final I2 package; N2 theory/exact-arithmetic closure	I2 where finite + theory audit
Short/return/long regime	N3–N5	authoritative N3, N4 and N5 I2 evidence packages	complete I2 replay
Complementary block	C0–C4	<code>R16_COMPLEMENTARY_TRUST_BASE_I2_EVIDENCE_20260926.zip</code>	complete I2 audit
Entry and pairing	U1–U2, A0–A2, E0–E1	<code>R16_FINAL_ENTRY_ASSEMBLY_I2_EVIDENCE_20260926.zip</code>	complete assembly audit
Seed-7 discharge	S2	final-entry audit + complementary trust-base audit	$14 + 2 = 16/16$
Final theorem	T0	dependency map, Claim–Evidence Matrix, trust-base map, nine-gate report	closed project level

8 Relation to previous work and scope

Joung and van Koert developed validated computational methods for symmetric periodic orbits and Conley–Zehnder indices in the PCR3BP (Joung and van Koert, 2025). Liu and Salomão prove, among other results,

Birkhoff’s retrograde-orbit conclusion for mass ratios sufficiently close to $1/2$ and all subcritical energies; the frozen source audit identifies the relevant all-subcritical result as Theorem 1.16 in the cited arXiv version (Liu and Salomão, 2025). The neighborhood in that result is existential for the comparison made here. We therefore do not claim that G459 lies outside their theorem.

This paper proves a fixed-mass all-subcritical theorem at $\gamma = 459/1000$. It does not prove an interval of mass ratios, optimality of the shear, uniqueness or nondegeneracy of all periodic orbits, or that every retrograde-looking numerical orbit is a valid rational-open-book binding. Historical VAST runs, broad orbit searches, failed global beta thresholds and exploratory floating scouts are provenance, not premises of the final proof.

9 Reproducibility and availability

The final release will contain the frozen dependency map, Claim–Evidence Matrix, trust-base map, exact parameter/environment manifests, theorem-critical proof objects, independent checkers, a quick-audit route, a complete replay route and SHA-256 manifests. Persistent repository and DOI identifiers will be inserted only after the release assets are frozen.

Statements and Declarations

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Author contributions. [To be completed by the author.]

Data availability. The theorem-critical finite certificates, manifests and audit objects will be deposited in the final public reproducibility release. [Persistent identifier to be inserted after deposit.]

Code availability. The proof-critical source, replay scripts and independent checkers will be released with the frozen reproducibility package. [Persistent identifier to be inserted after deposit.]

Use of AI-assisted tools. An AI language model was used during the research and publication workflow for organization, drafting support, code assistance, source comparison and audit support. AI output was not treated as theorem evidence. The theorem-critical computer-assisted claims are accepted only through deterministic exact/interval certificates, finite proof objects and independent replay/checkers. Scientific judgments, verification of the final mathematical claims and responsibility for the submitted text remain with the human author(s). The AI system is not an author.

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